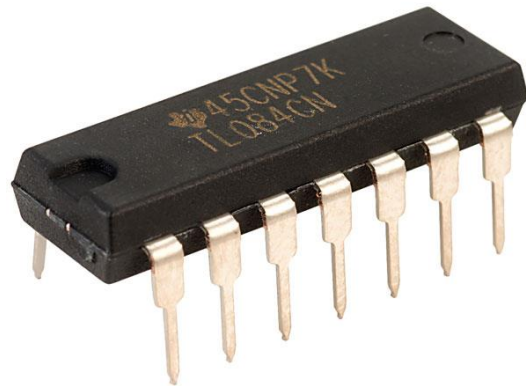


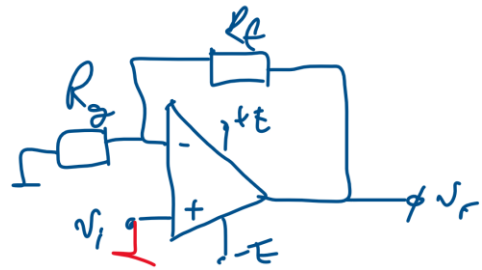
Bazele Circuitelor Electronice

Aplicații cu Amplificatoare Operaționale



Neidealități ale amplificatoarelor operaționale

Tensiunea de offset



$$R_g = 1 \text{ k}\Omega ; \pm E = \pm 15 \text{ V}$$

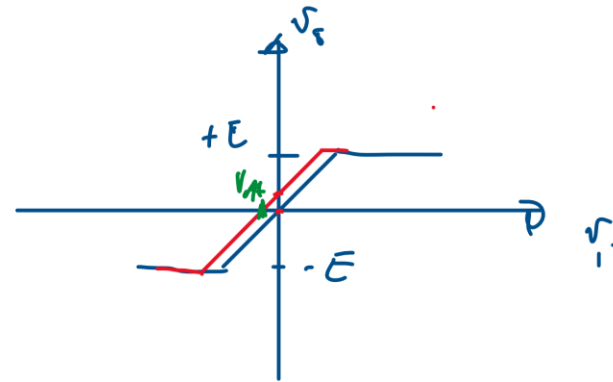
$$R_f = 33 \text{ k}\Omega$$

$$A = 1 + \frac{R_f}{R_g}$$

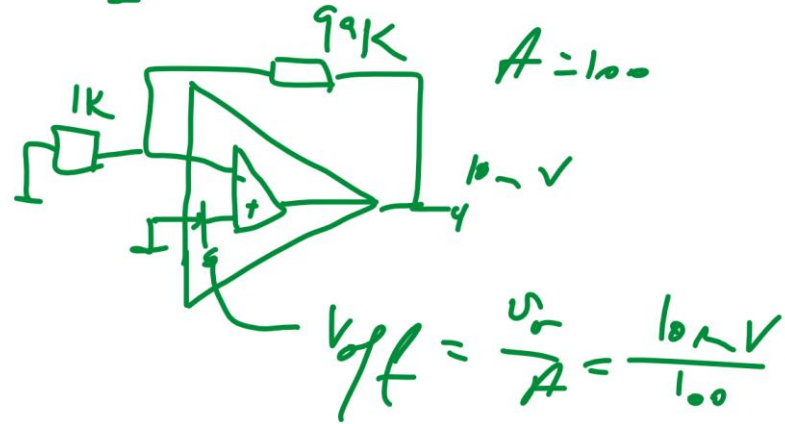
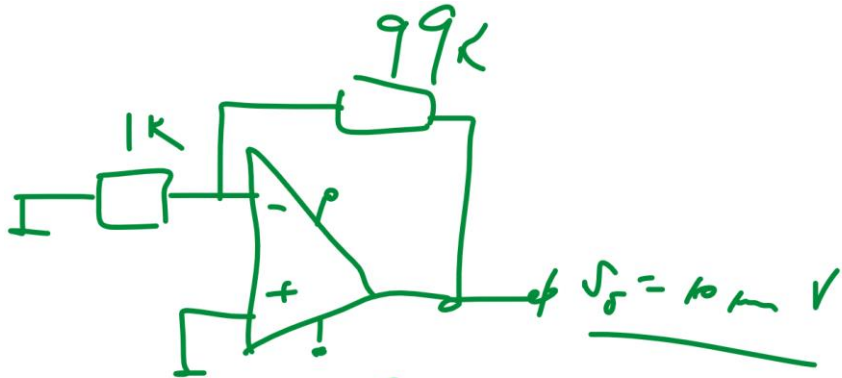
$$v_r = \left(1 + \frac{R_f}{R_g}\right) v_i = \left(1 + \frac{33 \text{ k}\Omega}{1 \text{ k}\Omega}\right) v_i$$

$$v_o = 100 v_i$$

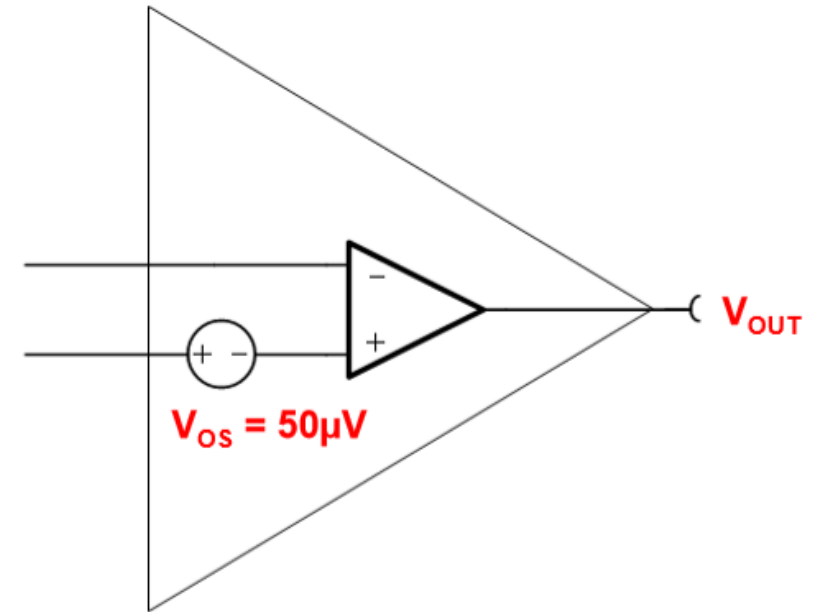
np. $v_i = 0 \text{ V}$ $\xrightarrow{A_o = \text{ideal}}$ 0 V



Tensiunea de offset



$$V_{off} = 0.1 \text{ mV} = \underline{100 \mu\text{V}}$$



Tensiunea de offset

LM741

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
Input offset voltage	$R_S \leq 10 \text{ k}\Omega$	$T_A = 25^\circ\text{C}$		1	5	mV
		$T_{AMIN} \leq T_A \leq T_{AMAX}$			6	mV
Input offset voltage adjustment range	$T_A = 25^\circ\text{C}, V_S = \pm 20 \text{ V}$			± 15		mV
Input offset current	$T_A = 25^\circ\text{C}$			20	200	nA
	$T_{AMIN} \leq T_A \leq T_{AMAX}$			85	500	
Input bias current	$T_A = 25^\circ\text{C}$			80	500	nA
	$T_{AMIN} \leq T_A \leq T_{AMAX}$				1.5	μA
Input resistance	$T_A = 25^\circ\text{C}, V_S = \pm 20 \text{ V}$		0.3	2		M Ω
Input voltage range	$T_{AMIN} \leq T_A \leq T_{AMAX}$		± 12	± 13		V
Large signal voltage gain	$V_S = \pm 15 \text{ V}, V_O = \pm 10 \text{ V}, R_L \geq 2 \text{ k}\Omega$	$T_A = 25^\circ\text{C}$	50	200		V/mV
		$T_{AMIN} \leq T_A \leq T_{AMAX}$	25			

Tensiunea de offset

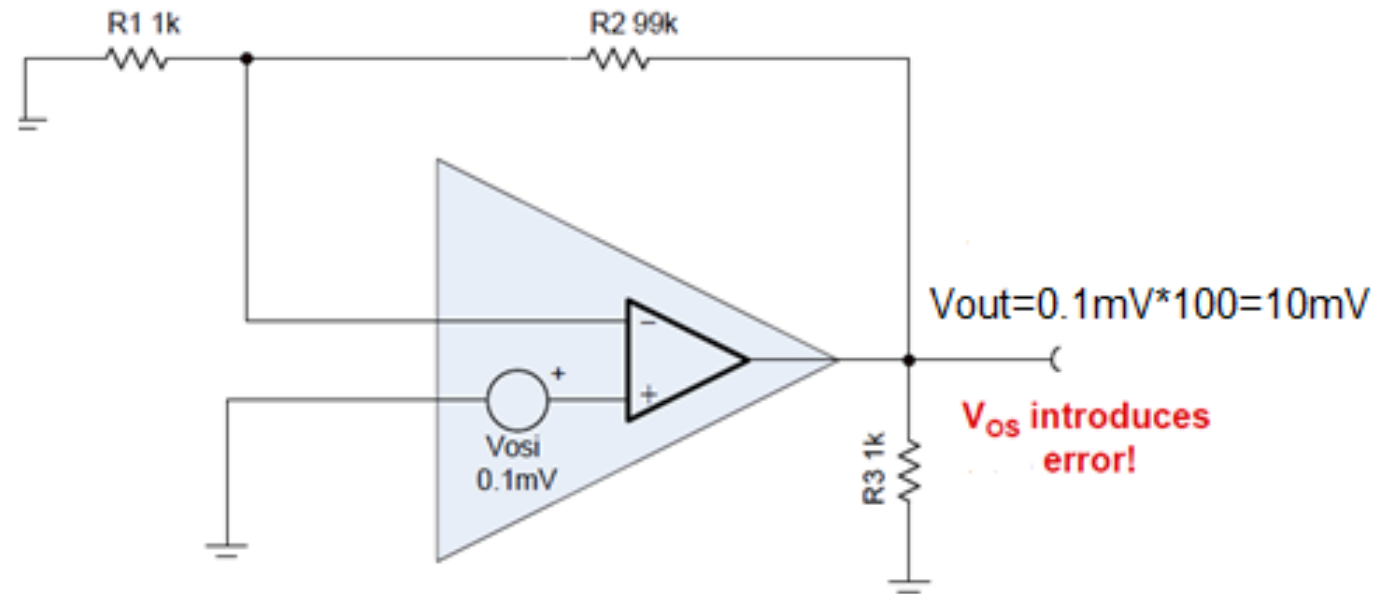
LM324

6.5 Electrical Characteristics for LMx24 and LM324K

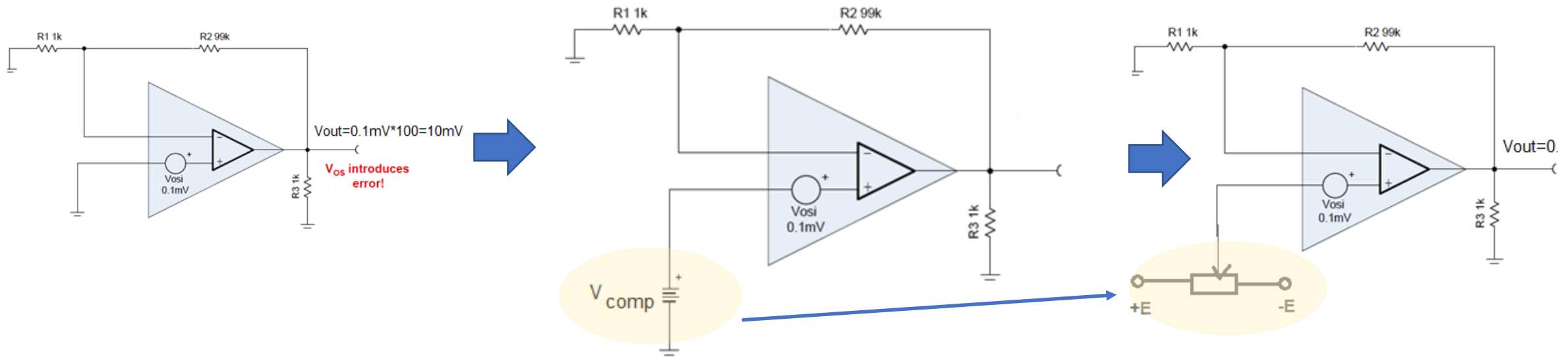
at specified free-air temperature, $V_{CC} = 5\text{ V}$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS ⁽¹⁾	T _A ⁽²⁾	LM124, LM224		LM324, LM324K		UNIT	
			MIN	TYP	MIN	TYP		
V _{IO} Input offset voltage	V _{CC} = 5 V to MAX, V _{IC} = V _{ICR} min, V _O = 1.4 V	25°C	3		3		mV	
		Full range	7		9			
I _{IO} Input offset current	V _O = 1.4 V	25°C	2		2		nA	
		Full range	100		150			
I _{IB} Input bias current	V _O = 1.4 V	25°C	−20		−20		nA	
		Full range	−300		−500			
V _{ICR} Common-mode input voltage range	V _{CC} = 5 V to MAX	25°C	0 to V _{CC} − 1.5		0 to V _{CC} − 1.5		V	
		Full range	0 to V _{CC} − 2		0 to V _{CC} − 2			
V _{OH} High-level output voltage	R _L = 2 kΩ	25°C	V _{CC} − 1.5		V _{CC} − 1.5		V	
	R _L = 10 kΩ	25°C						
	V _{CC} = MAX	R _L = 2 kΩ	Full range	26		26		
		R _L ≥ 10 kΩ	Full range	27	28	27		28
V _{OL} Low-level output voltage	R _L ≤ 10 kΩ	Full range	5		5		mV	
A _{VD} Large-signal differential voltage amplification	V _{CC} = 15 V, V _O = 1 V to 11 V, R _L ≥ 2 kΩ	25°C	50	100	25	100	V/mV	
		Full range	25		15			

Tensiunea de offset -COMPENSARE



Tensiunea de offset -COMPENSARE

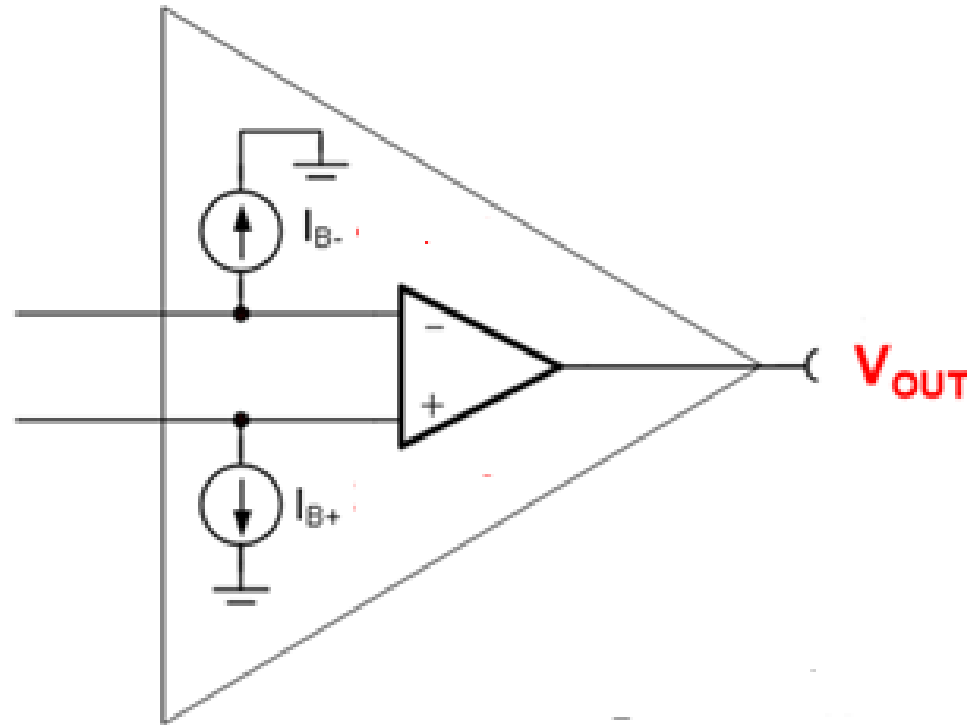


$$V_{comp} = -V_{os}$$

$$V_o = 100 \cdot (V_{comp} + V_{os}) = 0$$

Curenții de polarizare și curentul de offset

Curenții de polarizare și curențul de offset



Curenții de polarizare și curentul de offset

LM741

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
Input offset voltage	$R_S \leq 10 \text{ k}\Omega$	$T_A = 25^\circ\text{C}$		1	5	mV
		$T_{AMIN} \leq T_A \leq T_{AMAX}$			6	mV
Input offset voltage adjustment range	$T_A = 25^\circ\text{C}, V_S = \pm 20 \text{ V}$			± 15		mV
Input offset current	$T_A = 25^\circ\text{C}$			20	200	nA
	$T_{AMIN} \leq T_A \leq T_{AMAX}$			85	500	
Input bias current	$T_A = 25^\circ\text{C}$			80	500	nA
	$T_{AMIN} \leq T_A \leq T_{AMAX}$				1.5	μA
Input resistance	$T_A = 25^\circ\text{C}, V_S = \pm 20 \text{ V}$		0.3	2		M Ω
Input voltage range	$T_{AMIN} \leq T_A \leq T_{AMAX}$		± 12	± 13		V
Large signal voltage gain	$V_S = \pm 15 \text{ V}, V_O = \pm 10 \text{ V}, R_L \geq 2 \text{ k}\Omega$	$T_A = 25^\circ\text{C}$	50	200		V/mV
		$T_{AMIN} \leq T_A \leq T_{AMAX}$	25			

Curenții de polarizare și curentul de offset

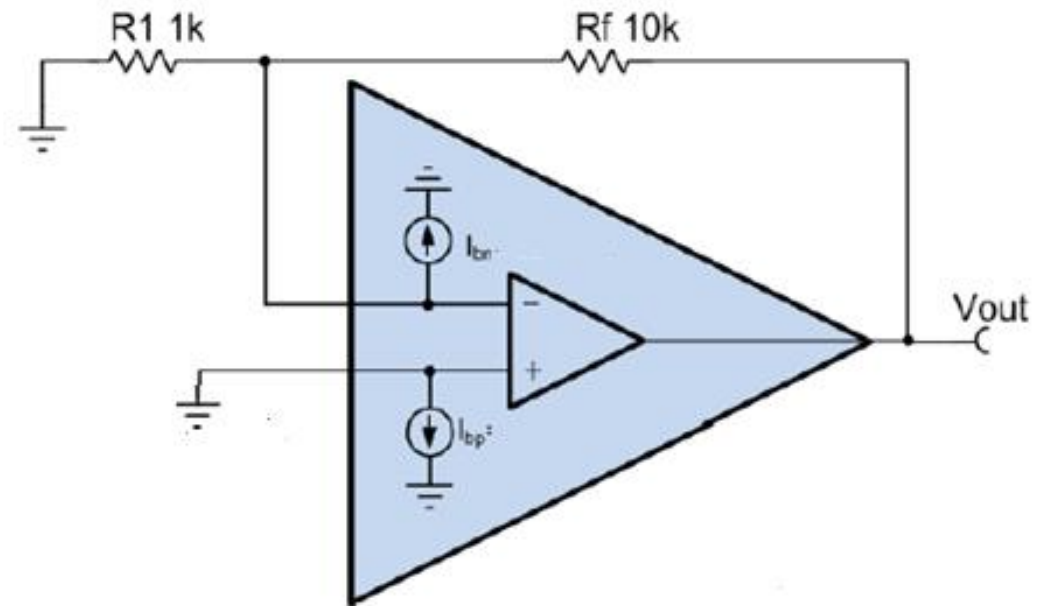
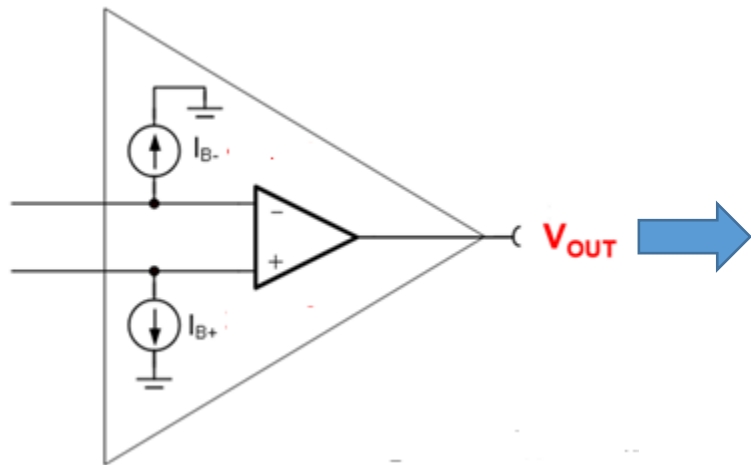
LM324

6.5 Electrical Characteristics for LMx24 and LM324K

at specified free-air temperature, $V_{CC} = 5\text{ V}$ (unless otherwise noted)

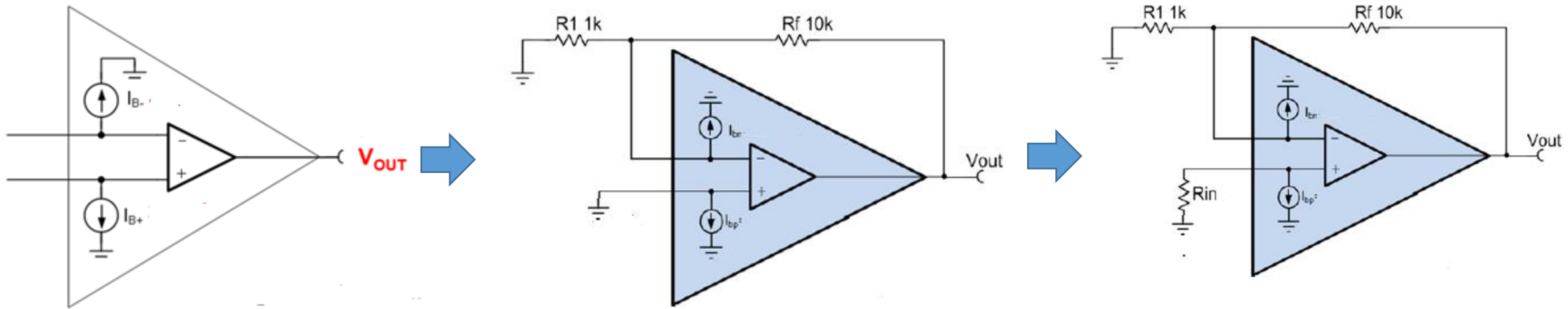
PARAMETER	TEST CONDITIONS ⁽¹⁾	T_A ⁽²⁾	LM124, LM224			LM324, LM324K			UNIT
			MIN	TYP ⁽³⁾	MAX	MIN	TYP ⁽³⁾	MAX	
V_{IO}	Input offset voltage	$V_{CC} = 5\text{ V to MAX}$, $V_{IC} = V_{ICRmin}$, $V_O = 1.4\text{ V}$		3	5		3	7	mV
I_{IO}	Input offset current	$V_O = 1.4\text{ V}$		2	30		2	50	nA
		Full range			100			150	
I_{IB}	Input bias current	$V_O = 1.4\text{ V}$		-20	-150		-20	-250	nA
		Full range			-300			-500	
V_{ICR}	Common-mode input voltage range	$V_{CC} = 5\text{ V to MAX}$		0 to			0 to		V
		Full range		$V_{CC} - 1.5$			$V_{CC} - 1.5$		
V_{OH}	High-level output voltage	$R_L = 2\text{ k}\Omega$							V
		$R_L = 10\text{ k}\Omega$							
		$V_{CC} = \text{MAX}$, $R_L = 2\text{ k}\Omega$		26			26		
		$V_{CC} = \text{MAX}$, $R_L \geq 10\text{ k}\Omega$		27	28		27	28	
V_{OL}	Low-level output voltage	$R_L \leq 10\text{ k}\Omega$		5	20		5	20	mV
A_{VD}	Large-signal differential voltage amplification	$V_{CC} = 15\text{ V}$, $V_O = 1\text{ V to } 11\text{ V}$, $R_L \geq 2\text{ k}\Omega$		50	100		25	100	V/mV
		Full range		25			15		

Curenții de polarizare și curențul de offset



$$v_{out} = A \cdot (v^+ - v^-) = A \cdot (I^- \cdot R_{ech} - 0) \neq 0$$

Curenții de polarizare și curențul de offset

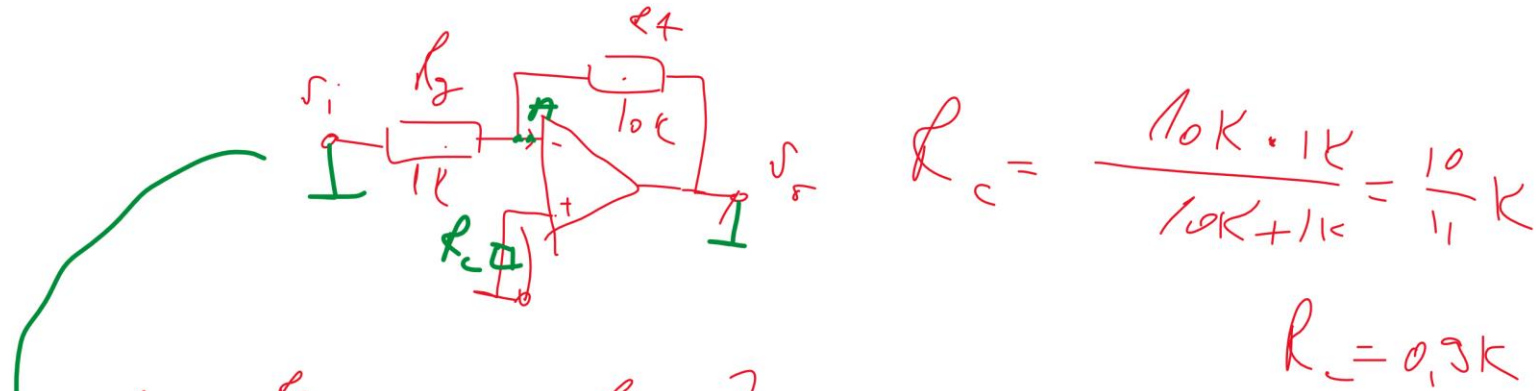


$$v_{out} = A \cdot (v^+ - v^-) = A \cdot (I^- \cdot R_{ech} - 0) \neq 0 \quad \Rightarrow$$

$$I^+ \cdot R_{in} = I^- \cdot R_{ech}$$

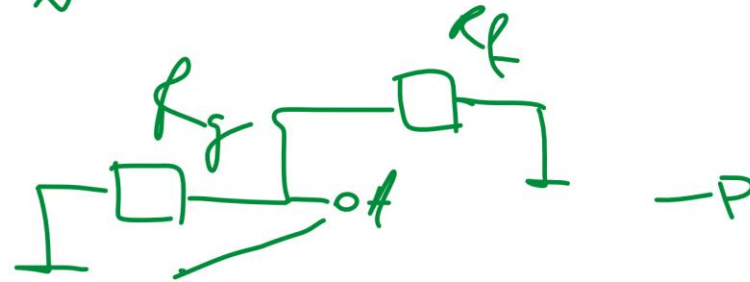
$$R_{in} = R_1 \parallel R_f$$

Curenții de polarizare și curențul de offset



$$A = -\frac{R_f}{R_g} \Rightarrow v_o = -\frac{R_f}{R_g} \cdot v_i$$

$v_i = 0$
 $\Rightarrow v_o \neq 0$



$$R_A = R_c = R_g \parallel R_f$$

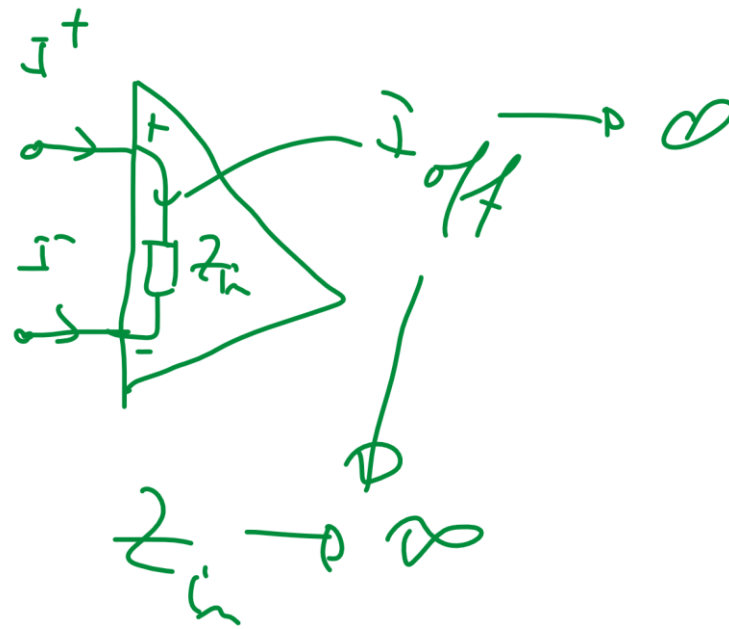
Curenții de polarizare și curentul de offset



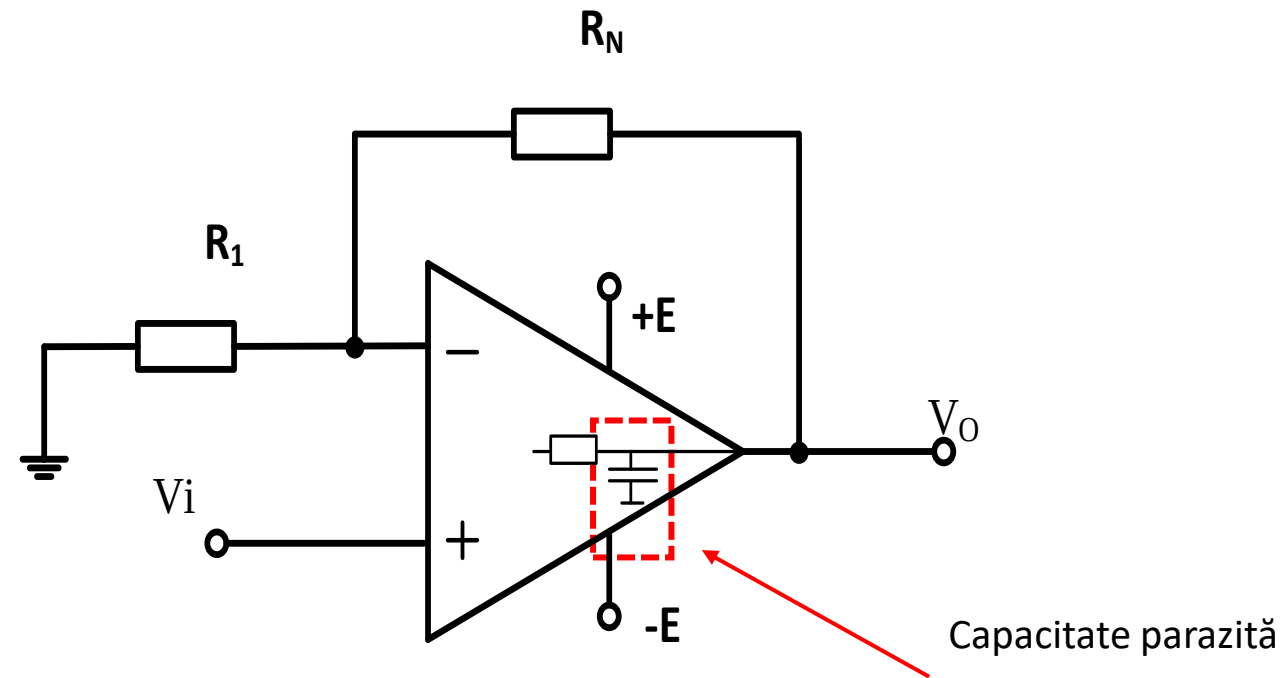
$$R_c = R_i \parallel R_f$$

Curenții de polarizare și curențul de offset

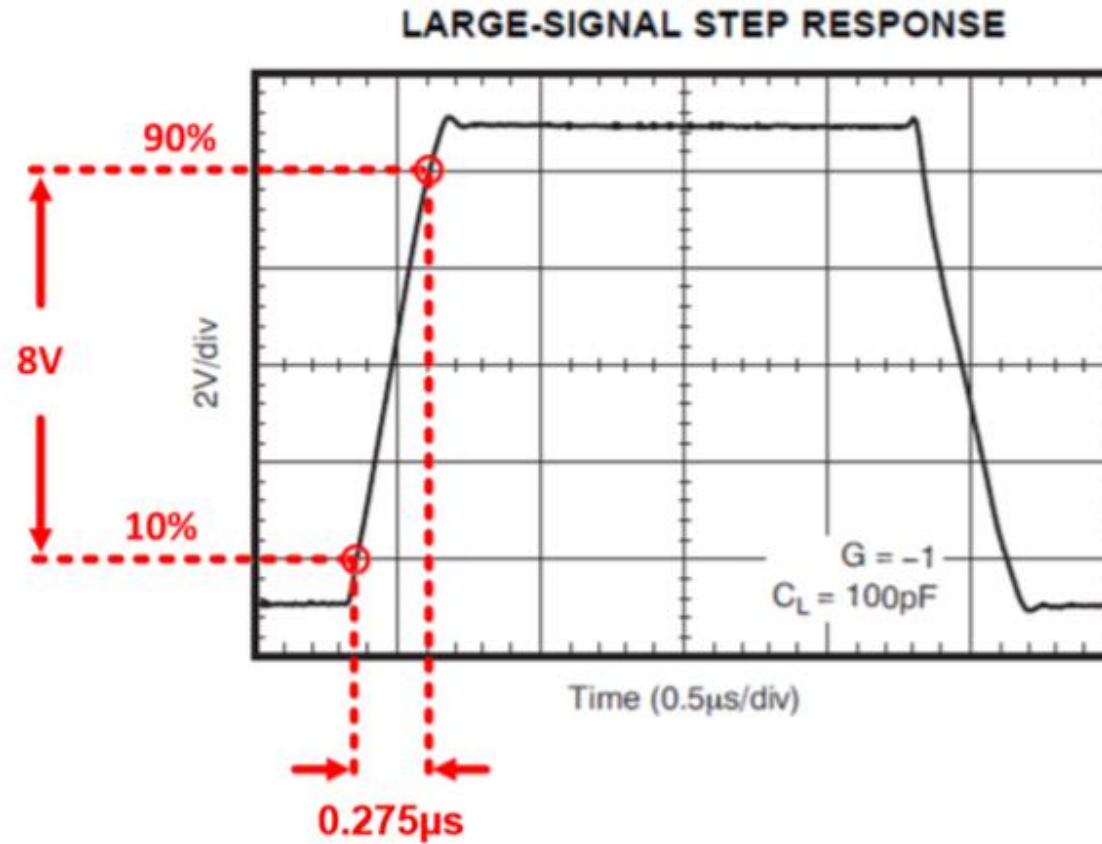
$$I_{off} = |I^+ - I^-|$$



Slew Rate

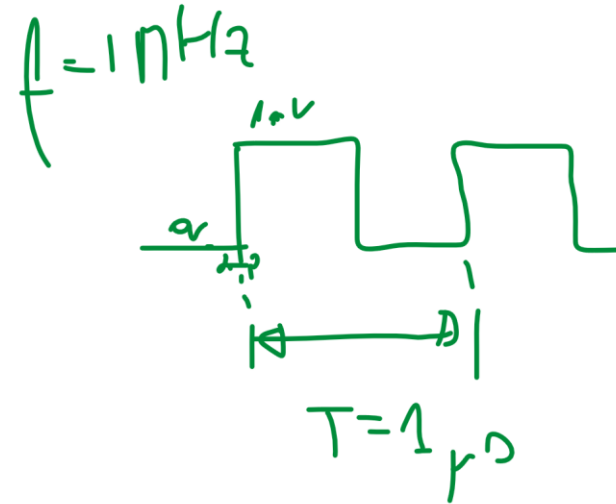


Slew Rate



$$\text{Slew_Rate} = \frac{\Delta V_{\text{out}}}{\Delta \text{Time}} = \frac{(V_{\text{out}90\%} - V_{\text{out}10\%})}{(t_{90\%} - t_{10\%})} = \frac{(9\text{V} - 1\text{V})}{(0.625\mu\text{s} - 0.35\mu\text{s})} = 29 \frac{\text{V}}{\mu\text{s}}$$

Slew Rate



LM741, $SR = 0.5 \text{ V}/\mu\text{s}$ 0.5V

TL081, $SR = 13 \text{ V}/\mu\text{s}$ ✓

Slew Rate

TL081

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
SR	Slew rate at unity gain	$V_I = 10\text{ V}$, $R_L = 2\text{ k}\Omega$, $C_L = 100\text{ pF}$, See Figure 19	$8^{(1)}$	13		V/ μs
		$V_I = 10\text{ V}$, $R_L = 2\text{ k}\Omega$, $C_L = 100\text{ pF}$, $T_A = -55^\circ\text{C}$ to 125°C , See Figure 19	$5^{(1)}$			

LM741

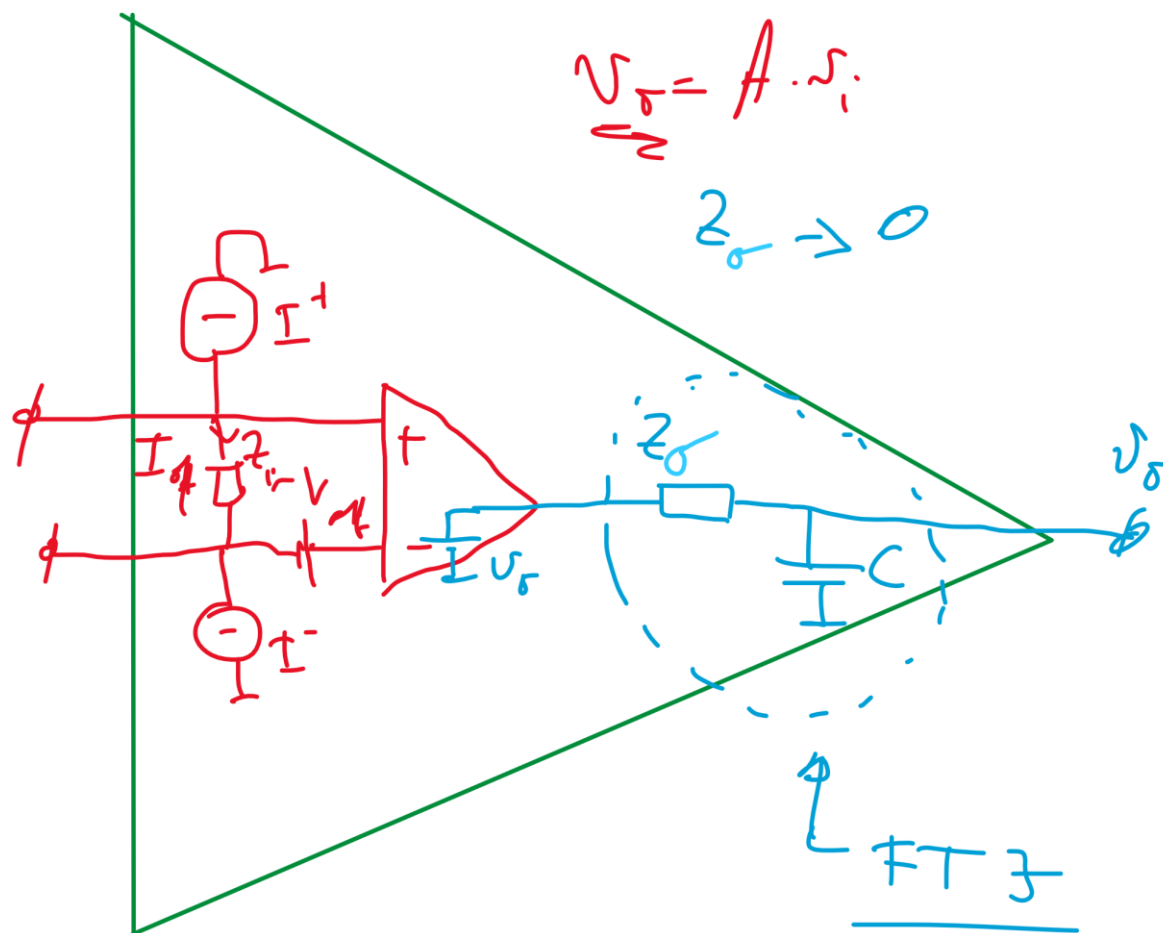
Slew rate	$T_A = 25^\circ\text{C}$, unity gain	0.5	V/ μs
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Banda de frecvențe

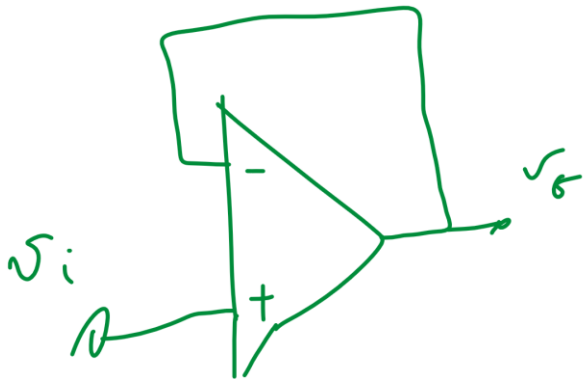
	PARAMETER	TEST CONDITIONS ⁽¹⁾	T _A	TL081M, TL082M			TL084Q, TL084M			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
A _v	voltage amplification	V _{OL} = 0V, R _L = 10kΩ	Full range	15			15			
B ₁	Unity-gain bandwidth		25°C		3			3		MHz
r _i	Input resistance		25°C		10 ¹²			10 ¹²		Ω

$$B_{f(A_x)} = \frac{B_{f(A=1)}}{A_x}$$

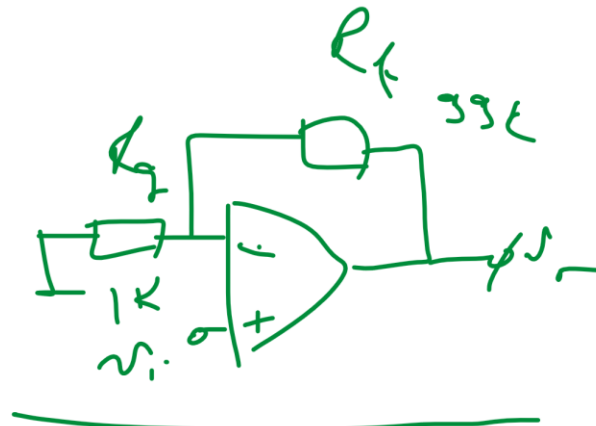
Banda de frecvențe



Banda de frecvențe



$$\frac{A=1}{v_o = v_i}$$



$$\frac{TL084}{4} \quad f_{max} = 3 \text{ MHz}$$

$$A = 100$$

$$f_{max} = 0.03 \text{ MHz} = 30 \text{ kHz}$$

